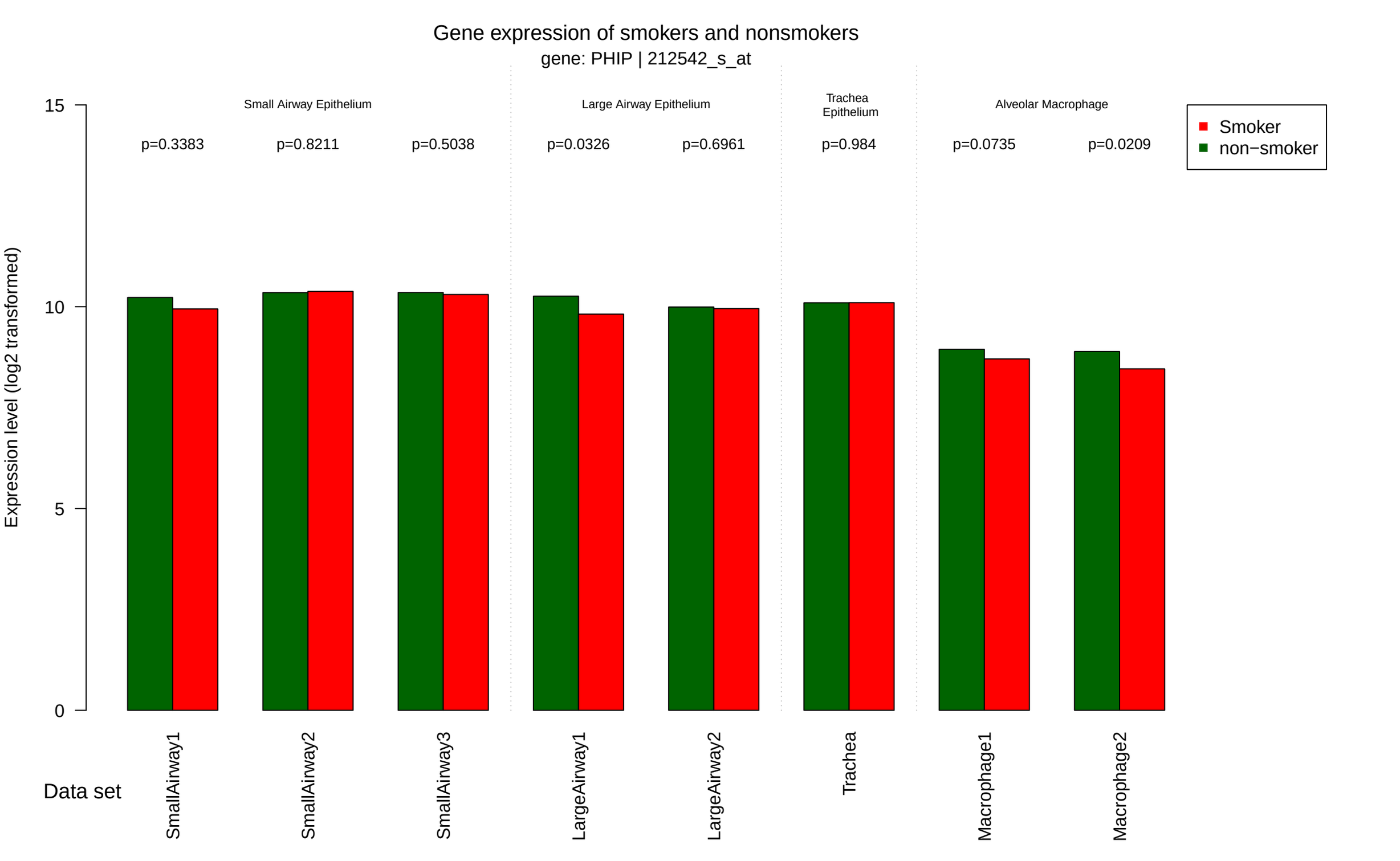
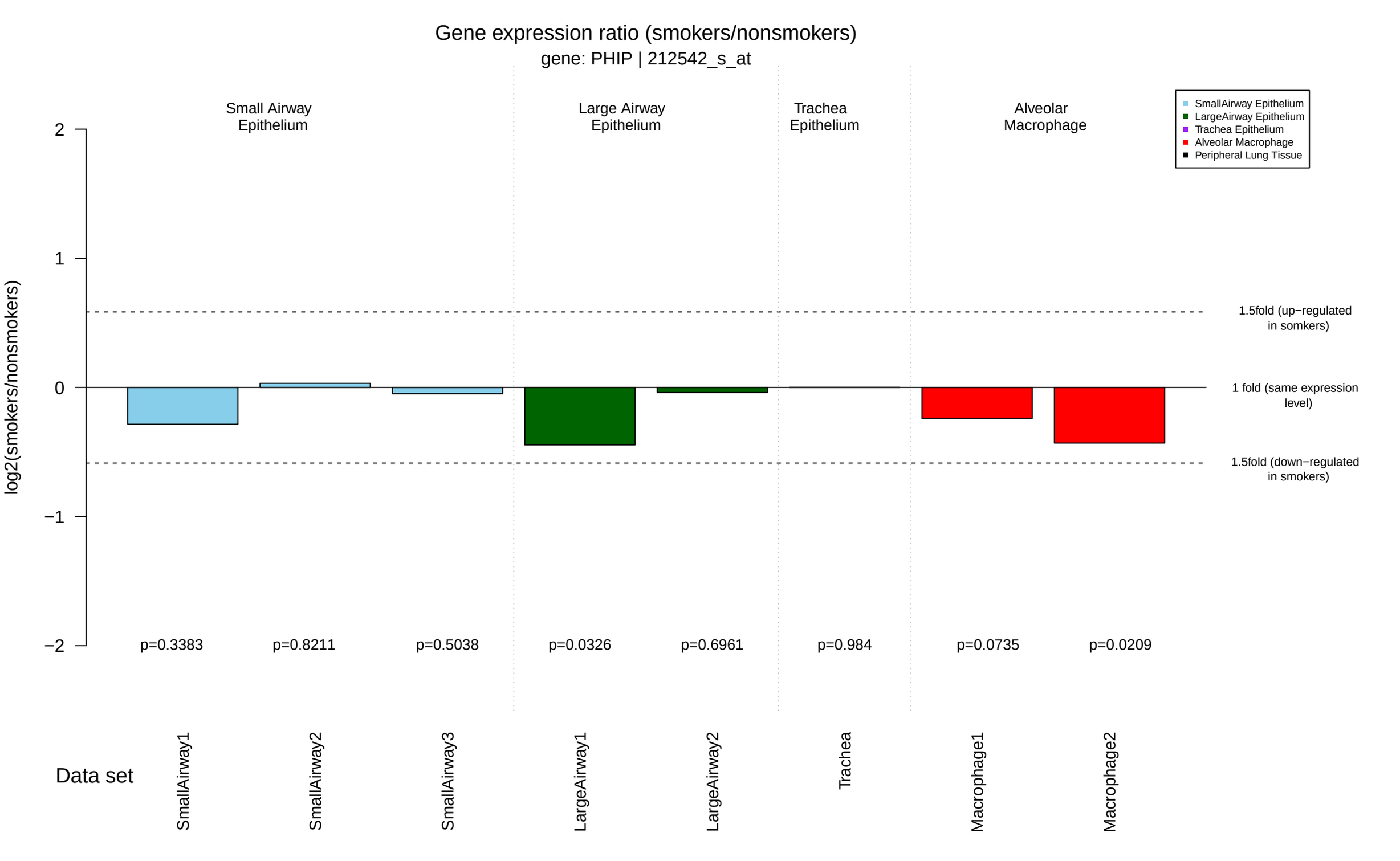
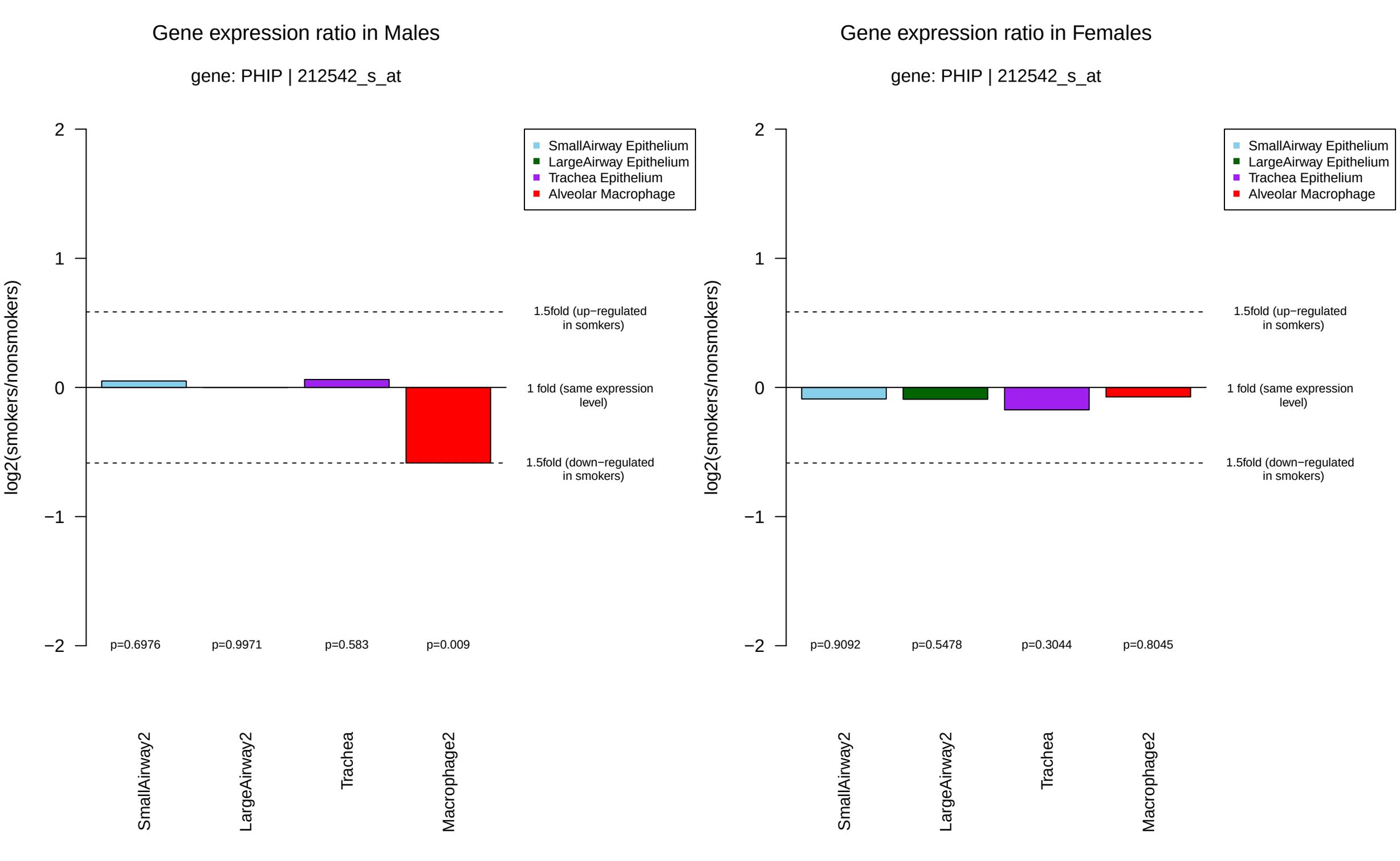
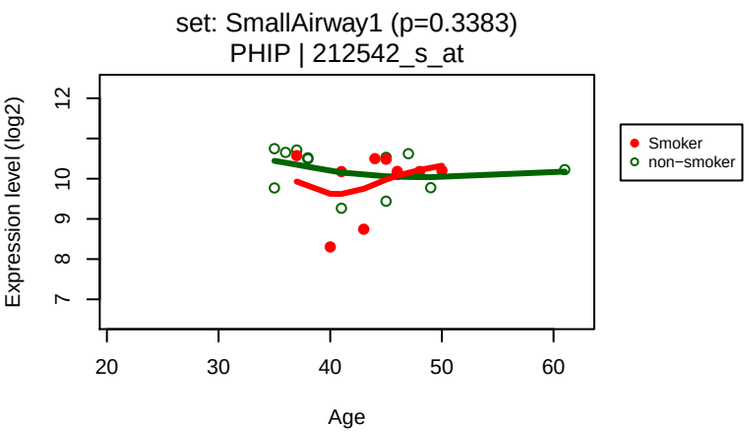
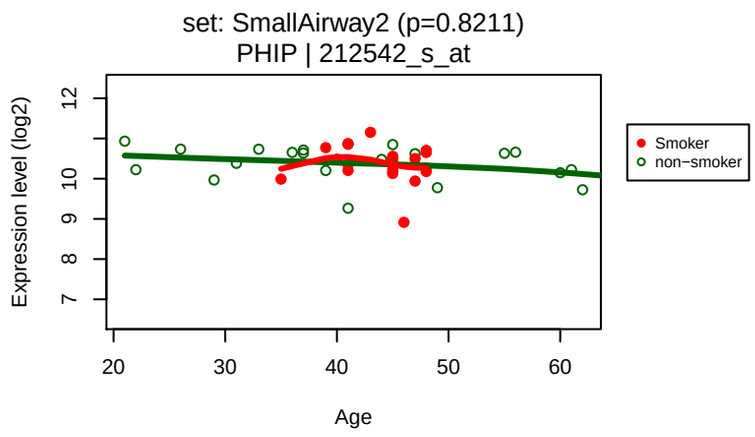
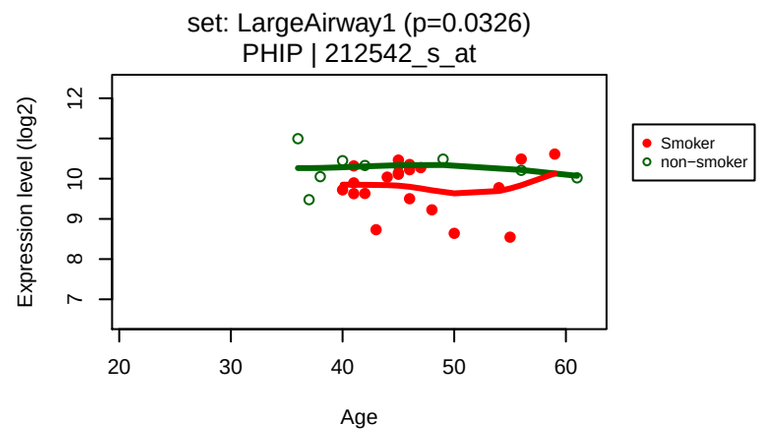

Gene expression of smokers and nonsmokers
gene: PHIP | 212542_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3383
p=0.8211
p=0.5038
p=0.0326
p=0.6961
p=0.984
p=0.0735
p=0.0209
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

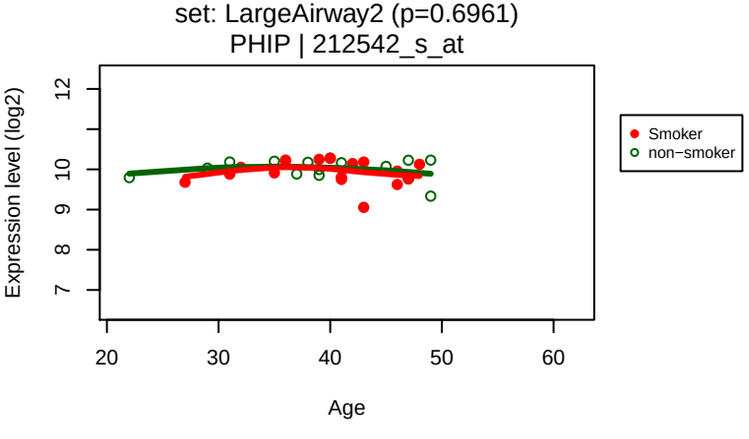

Gene expression ratio (smokers/nonsmokers)
gene: PHIP | 212542_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3383
p=0.8211
p=0.5038
p=0.0326
p=0.6961
p=0.984
p=0.0735
p=0.0209
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

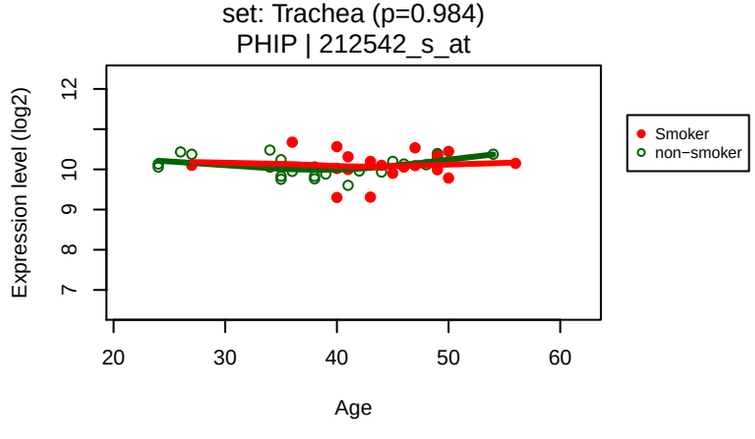

Gene expression ratio in Males
gene: PHIP | 212542_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6976
p=0.9971
p=0.583
p=0.009
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PHIP | 212542_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9092
p=0.5478
p=0.3044
p=0.8045
SmallAirway2
LargeAirway2
Trachea
Macrophage2

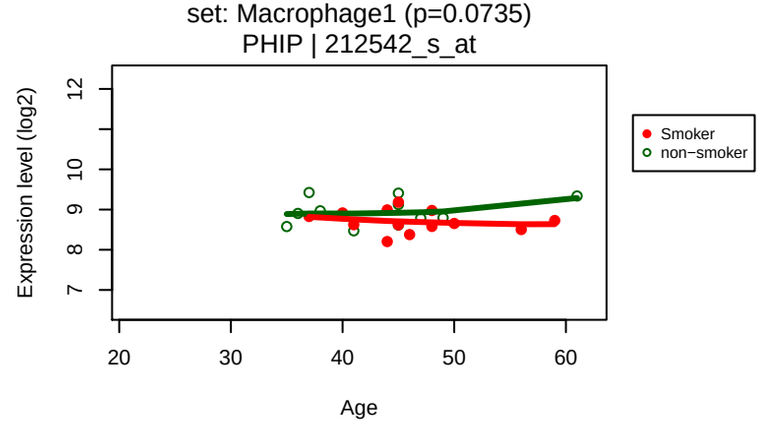

set: SmallAirway1 (p=0.3383)
PHIP | 212542_s_at
Expression level (log2)
Age
Smoker
non−smoker

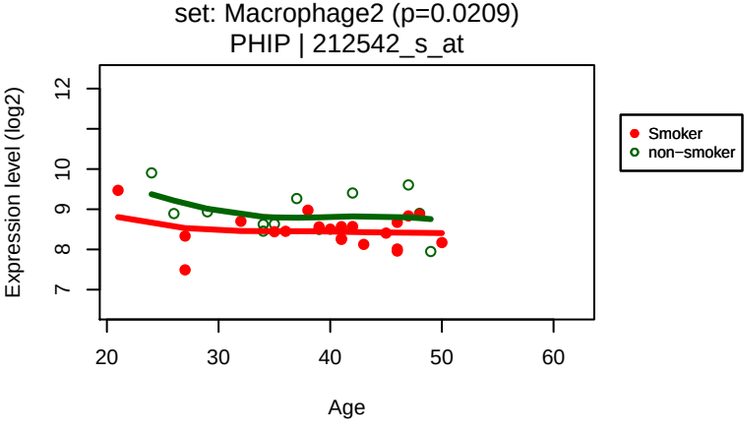

set: SmallAirway2 (p=0.8211)
PHIP | 212542_s_at
Expression level (log2)
Age
Smoker
non−smoker

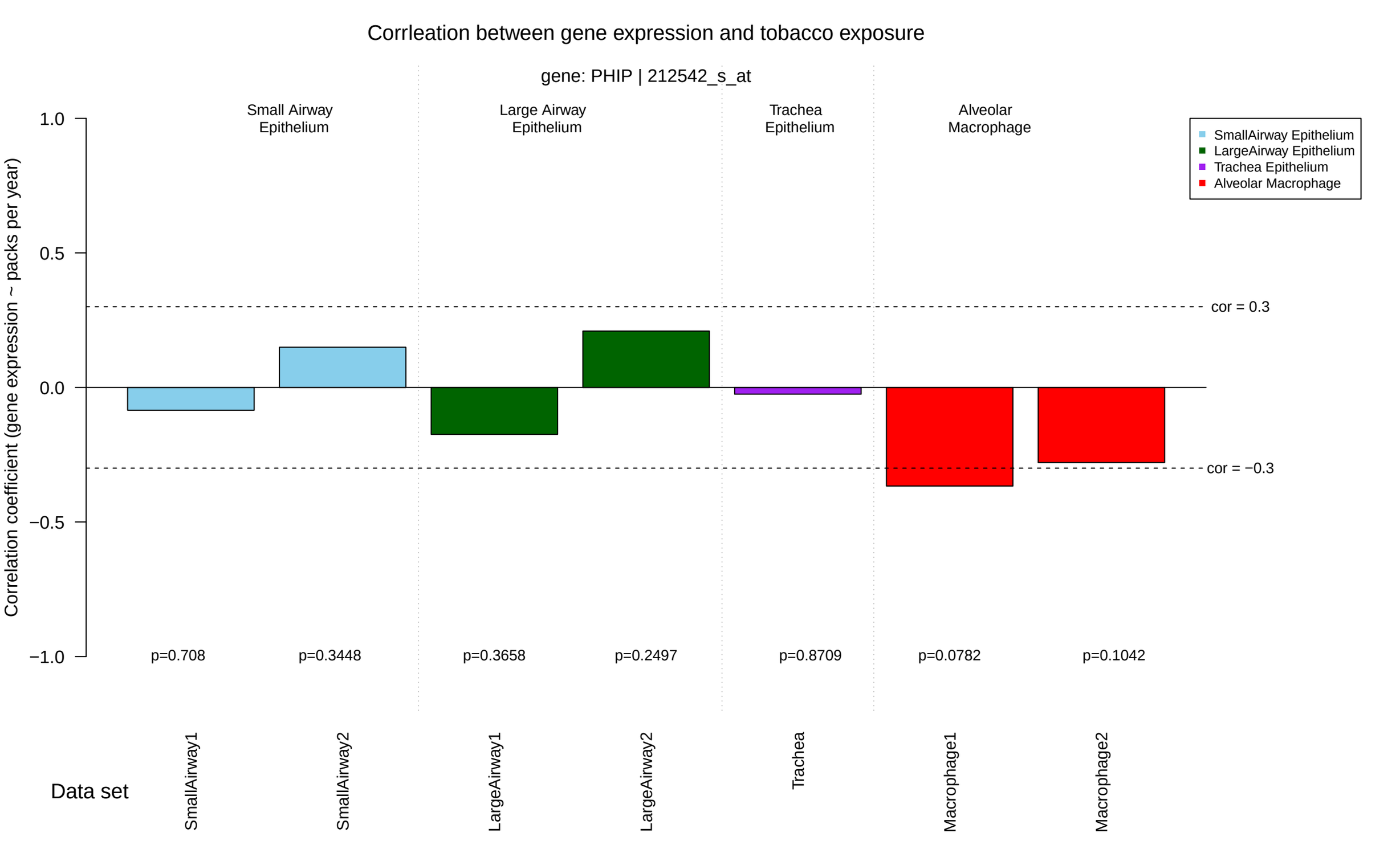

set: LargeAirway1 (p=0.0326)
PHIP | 212542_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6961)
PHIP | 212542_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.984)
PHIP | 212542_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0735)
PHIP | 212542_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0209)
PHIP | 212542_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PHIP | 212542_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.708
p=0.3448
p=0.3658
p=0.2497
p=0.8709
p=0.0782
p=0.1042
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2